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Children and Stress: The new science on chronically harsh and conflict-ridden households.

By Alan E. Kazdin and Carlo Rotella Posted Wednesday, Aug. 11, 2010, at 10:06 AM ET



You work and work to provide for your kids, and that puts you under a lot of stress. Job insecurity, maybe some mortgage problems, and other common afflictions of the times only increase the pressure. You find yourself taking it out on your family—the kids, your spouse, the cat, any unfortunate who gets in your way. When you're home and not obsessively checking your e-mail, you lose your temper, you snap and yell and brood, you run alternately too hot (angry and aggressive, spoiling for a fight) and too cold (withdrawn and distant, a forbidding stone-face). You'll admit that you're hard to be around, but look, life is tough and you're knocking yourself out without much in the way of thanks or respite to make enough money to feed and house the kids, and that's what matters most, right?

Well, yes and no. Providing for children's basic material needs is essential—it would be silly to argue otherwise. But a chronically harsh, conflict-ridden, chaotic household environment can do psychological damage and related *physical* damage that undercuts the good effects of whatever you're doing to provide for children's basic needs. What researchers are finding out about the effect of early stress on children's long-term health may well cause you to rethink your priorities. So far, the developing body of research in this area tells us at least this much:

- Early stress in the home is significantly associated with later physical disease and early death. We'll get into the specifics shortly, but the diseases in question run across a wide range, from cancer to heart and liver and lung disorders to reproductive health problems to higher rates of alcohol abuse and smoking.

- Extremes such as child abuse increase risk (see Sidebar 1) for these outcomes, but less extreme influences predict these outcomes as well. We're talking about sustained conflict, a lack of emotional warmth in family relations, inadequate and inept parenting practices (e.g., heavy use of reprimands, punishments, shouting), and general chaos and disruption.

- For a harsh stressful environment, there is a dose-response relation: As stress rises higher, so do deleterious neuroendocrine changes in the child's body that relate to susceptibility to disease.

- The real danger comes from chronic stress—the body's adaptive reaction to stress that goes off and on as part of ordinary life does not seem to have undesirable consequences.

- Because not everyone responds the same way to chronic stress, we can say only that harsh environments increase risk, not that they invariably produce bad outcomes.

Before we discuss the research and its implications, we want to underscore a couple of points. First, don't misread what follows as an incitement to panic because you argue with your spouse or yell at your kids now and then. Most people with spouses or kids do that. We're talking about sustained, ongoing stress, not the usual ups and downs of family life. Second, what follows is not an attack on strict parenting. From the point of view of the science, a lot of order or not so much order can both be OK or not OK, depending on what else is going on in a household. Psychologists have known for decades that children exposed to harsh environments are at increased risk for all sorts of negative psychological consequences. The social, emotional, behavioral, and cognitive damage piles up over the short and long terms. Children who are abused are more likely to have higher rates of alcohol and drug

abuse, anxiety, depression, stress disorder, eating and sleep disorders, hyperactivity, impairments in thinking, suicidal behavior and self-harm, poor relationships with others, and low self-esteem. They are more likely to do poorly in school, be kept behind in grade level, and go less far in their education.

We also know that not everybody who was abused will show the same outcomes. Subtle variations in genes, for instance, help regulate chemical reactions in the body—the activity of hormones and neurotransmitters—when it responds to stressful events like abuse. A particular set of stressors may overactivate one child's system, leading to depression, but affect another child to a much lesser degree.

We've also learned more in recent years about *how* harsh environments in childhood can increase the risk for physical illness and early death. This isn't limited to the immediate physical effects of abuse: the brain and abdominal injuries, bruises and welts, burns and scalds, fractures, lacerations. Set those aside for the moment, although we realize that's a lot to set aside: The widespread tendency around the world to ignore, cover up, or misidentify the causes of children's deaths makes it hard to arrive at defensible numbers, but a World Health Organization study of violence estimated that approximately 57,000 children in the world per year suffer fatal abuse (see Sidebar 2) with the greatest risk for children who are 4 or younger.

When the vast majority of abused children who survive the immediate abuse grow up, they are at increased risk for cancer, chronic lung disease, heart and liver disease, reproductive health problems, sexual dysfunction, and sexually transmitted diseases. Among the many studies in support of this conclusion is the Adverse Childhood Experience Study, an ongoing project that has examined the medical histories of approximately 17,000 adults to determine that those who had been exposed to neglect, abuse, or family violence as children were 1½ to 2 times more likely to die prematurely, have heart disease, or suffer autoimmune disorders.

It's important, however, not to confine this discussion to situations that answer to the legal definition of child abuse. We need that legal definition so we can track patterns and so government can decide when to intervene in family life, but psychologists are coming to see a great deal of harm resulting from mistreatment of children that does not rise to the extreme levels that qualify legally as abuse. Harsh and chaotic early environments that are not labeled as physically abusive do increase the risk of adverse psychological consequences.

Here's where the science is producing fresh news. Recent work in this area points to three main conclusions.

First, less-extreme harsh environments still lead to physical health problems. Children don't have to be abused in a strict legal sense to suffer lasting harm.

Second, such harsh environments can be produced by multiple factors that individually don't seem that significant but can influence one another in a snowball effect. For example, low-level conflict between parents that continues a bit longer than usual can lead to changes in how they talk to their child, and a little more edge and stress in the parent's voice can increase child misbehavior, such as more noncompliance. That push-back from the child grates with special force on a parent's nerves that are already raw because of some added problem, like real or threatened loss of a job.

Everyone's reactions go up a notch, with raging parents and whiny children egging each other on in an ascending spiral of stress. Such things normally happen in most homes without long-range effect, but they are usually short-lived. When the stressors continue for weeks and months, are a little more intense than usual, and seem to become a way of life, significant psychological and physical problems are more likely to emerge. No one knows exactly how much stress will do it. Even if we did identify a tipping point, it would vary for each family and individual based on many other influences—such as, for instance, other sources of emotional support in each person's life.

Third, researchers have made progress in understanding the precise nature of the connections between the psychological conditions in which a child is raised and the heightened risk of disease later in life and early death. One view has been that growing up in harsh environments makes a person more likely to engage in risky behaviors such as smoking, excessive drinking, and risky sexual practices, thereby make children from such homes more likely to have some of the diseases mentioned previously. Maybe these kids grow up to be less healthy just because they do more things that we already know are bad for you. But even if that's true, it's not the whole story. The research shows that stress reactions and hormone and neuroendocrine responses all by themselves, independent of risky habits like smoking or heavy drinking, are different among people with early exposure to stress.

Some of the recent research along this line has been done on animals, which often have biological characteristics similar to ours but can be studied under controlled conditions that are difficult or impossible to achieve with human subjects. For example, exposing mice to chronically stressful living conditions by repeatedly introducing an aggressive mouse intruder into their cages disrupts their social hierarchy. For the mice, this is roughly equivalent to the effect on us of having a representative from a local gang or the IRS visiting us at dinnertime each night to go over our finances in depth. The mice living in the cage show signs of fear, submissiveness, and stress, leading to biological changes that harm the ability of their immune systems to deal with disease. These changes are very much like what is found in the human studies, and yet the mice don't smoke and drink to deal with their stress.

Human studies have also been conducted to chart the course of biological responses to stress. They show in more detail the physiological changes that control activation of the immune system and that control inflammation, a characteristic result of assaults on the body from bacteria, virus, or injury. Inflammation in various organs and the circulatory systems contributes to age-related medical conditions such as heart disease, cancers, Alzheimer's, diabetes, metabolic syndrome, autoimmune disorders, and more. The research is showing us that, under chronic stress, a complex biological reaction kicks in that affects the neuroendocrine system and that these changes—independent of behaviors that might impair health—increase a person's susceptibility to disease by rendering the immune system less vigilant and effective.

Again, not everyone is affected by stress to the same degree, but we can already say, in a general way, that the findings oblige us to reconsider some conventional wisdom.

The prime example is "What does not kill you will make you stronger." That's true often enough in life, especially if you're a serious weightlifter, but in relation to stressors it's not. Stressors that do not kill a child in the short run can weaken the body and lead to a lot of illness, and they can kill. The more accurate amended cliché would be "What does not kill you can sometimes make you stronger, but it can definitely make you weaker too." Or, more simply, "What does not kill you now may kill you later."

These are findings that families can try to do something about. Being aware that sustained stress and conflict in the home have silent effects is important. We have learned from public awareness campaigns that high blood pressure is "the silent killer." High ongoing stress in the home for a child is similarly dangerous. A parent will never see the changes in the immune system of a child that have been documented in human and animal studies, and routine laboratory blood tests, urinalyses, and camp physicals do not test for such changes. A parent may not be able to tell if a child has more colds or illness when compared to other children. But the research shows us that for many people the negative effects are there, hidden from casual view, and will continue to show up decades after the child leaves the home.

What can one do? Some of the modest, daily measures we have discussed in other articles take on special importance. Activities that can bring stability to home life operate to reduce stress. Routines and rituals—going to the store together, playing a game, engaging in some regular family activity, eating meals together—help reduce stress. Family activities, assuming they are not riddled with conflict, are all to the good. These activities are not necessarily about fun, although fun is fine, and they're not about spending money. Downtime, family time, and everyday routines all can alleviate stress. Recommending them is way beyond common sense now: They are an investment in physical health and are "psychological nutrients" for a sound immune system.

And, of course, you can take a kinder tone with your kids, no matter how strict you may be with them. A recent study has shown that reducing harsh parenting—draconian discipline, sharp criticism—and increasing warmth, affection, and praise for good behavior contributed to reduction in biological indices of stress responses, like the levels of cortisone, a stress hormone. Again, this doesn't mean that anyone's advocating that parents become permissive or weak. The opposite, in fact: Going off on your kids because you're frustrated is a form of giving in to weakness; it takes a lot more strength to be gentle with them even while exercising your authority and coping with your own undeniably stress-inducing problems.

The current recession has made it especially timely to be sensitive to stressors of family life. The Foundation for Child Development recently evaluated current research and its impact on children and found that more children in this country are living in poverty now than in the past 20 years, with approximately 20 percent of youth living below the poverty line, and more are living in homes in which parents do not have secure employment. Marital discord and conflict in the home are heavily affected by economic conditions that can lead to, for instance, reduced health care or a poorer diet. More than ever, it's important to recognize that chronic household harshness and chaos can do lasting harm to children even when it doesn't rise to the level of legally defined abuse.

Here is a list of resources and further reading. You can find a wealth of scientific information there, but you might also stop and think for a moment about the plain-and-simple implications of what it tells us. As disturbing as it might be to learn that harshness in the home can damage your child's long-term health, it's not like the antidote is mysterious or expensive. Think of whatever extra warmth, affection, and other stress-minimizing kindness you can give to your child as deposits in his or her psychological bank account that will bear dividends by improving the chances of a long, healthy life.

Resources

Miller. G.E., & Chen, E. (April, 2010). Harsh family climate in early life presages the emergence of a proinflammatory phenotype in adolescence. *Psychological Science*.
World Report on Violence and Health. 2002; Geneva: World Health Organization.
The World Health Report 2002: Reducing Risks, Promoting Health Life. 2002. Geneva: World Health Organization.
www.aboutourkids.org/articles/csc_prevention_research_news_update_early_family_intervention

[Sidebar 1]

What “increased risk” means

“Risk” is a scientific term that means an individual who is exposed to some condition is more likely to experience some outcome later, when compared to someone who has not been exposed to that prior condition. Increased risk does not necessarily mean that the outcome is determined or that the outcome invariably will occur. The concept is very useful in science in understanding and preventing disease, but it can be frustrating and widely misunderstood in everyday life.

For example, when we first hear of a risk factor (say, cigarette smoking) and outcome (early death from lung cancer) our mind naturally goes to exceptions. Look at Grandma, who smoked camels for 20 years and then switched to cigarettes for 20 more years, retired from rodeo competition at 80, and lived to be 120; and look at Aunt Phyllis, who never smoked and avoided foreign films in which people smoke a lot, and she died of lung cancer at 40. To the public these exceptions serve to negate the rule or make science look silly.

We also tend to want simple rules: x causes y. Once we know the rule, we feel that we can do something about the threat it addresses. Bites from rabid dogs and raccoons cause rabies, and submersion in cold water leads to hypothermia and potential death. These are rules we can grab onto and use to map clear courses of action: Avoid dogs and raccoons who are muttering to themselves and foaming at the mouth, especially if they’re in really cold water. But there are relatively few clear-cut cases like this in science. For many medical and psychological conditions, there are multiple causes of varying importance. For example, high cholesterol is a risk factor for heart disease, and reducing cholesterol reduces the likelihood of heart disease. That looks like a rule of stay-clear-of-rabid-dogs-like clarity until we learn that about 50% of the people who die of heart attack had perfectly fine cholesterol.

But increased risk is serious business, and scientific understanding of it, even when it doesn’t yield simple cause-effect rules of action, saves lives on a daily basis in relation to cancer, heart disease, HIV, and more.

[Sidebar 2]

Definitions of Abuse

There is no single definition of abuse. In the United States variations of abuse—physical (heavy use of corporal punishment), neglect (deprivation of food or medical care, poor hygiene, abandonment, inadequate protection from danger, inadequate supervision), sexual abuse—are defined by state rather than by federal law. Internationally, different nations and cultures vary in their definitions. One general standard used by the World Health Organization to identify commonalities defines child abuse as “all forms of physical and/or emotional ill-treatment, sexual abuse, neglect or negligent treatment or commercial or other exploitation, resulting in actual or potential harm to the child’s health, survival, development, or dignity . . . .” But this definition leaves a lot of gray area, especially when poverty, war, and other such conditions make it difficult for adults to care for children.

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Photograph of a stressful household by Goodshoot.